Kartik Rajan Neralwar

Curriculum Vitae

Education

07/2021–Present **Ph.D. student**, International Max Planck Research School (IMPRS), *Universität Bonn, research performed at Max Planck Institute for Radio Astronomy (MPIfR)*, Germany
Thesis: "A multiscale study of molecular gas in STARFORGE"; advisors: Dr. Dario Colombo; Prof. Dr. Karl Menten (late), Prof. Dr. Pavel Kroupa, Dr. Daniel Seifried; expected submission in November 2025.

09/2018–01/2021 **Master of Science (Astrophysics)**, *Universität Bonn*, Germany
Thesis: "The SEDIGISM survey: Morphology of molecular clouds"; advisors: Dr. Dario Colombo; Prof. Dr. Karl Menten, Prof. Dr. Pavel Kroupa.

07/2015–06/2018 **Bachelor of Science (specialisation in Physics)**, *Fergusson College*, Pune, India
Thesis: "Finding observational proxies for the forces acting during the initiation stage of Coronal Mass Ejections (CMEs)"; advisors: Dr. K. Sasikumar Raja and Prof. Dr. Prasad Subramanian at IISER Pune

Teaching and Tutoring

06/2024–09/2024 **Co-organiser and speaker**, Deep learning in Astrophysics *workshop series*, Regionales Rechenzentrum der Universität zu Köln, Germany
Co-organised two workshops on Deep Learning in Astrophysics and gave lectures on the topics "Introduction to Deep Learning" and "A simplified view of neural networks".

09/2023–02/2024 **Tutor**, Statistical methods for astrophysics and cosmology, University of Bonn, Germany
Tutored the course astro8506: Statistical methods for astrophysics and cosmology for M.Sc. Astrophysics.

05/2016–04/2017 **Fellow**, Science Education Initiative (SEI), Pune, India
Taught science and mathematics to students in grade 5 in a government school in Pune.
Trained fellow inductees at Science Education Initiative (SEI) in basic physics.

Supervision of students

Internship	*"CASI-3D U-Net for ^{13}CO Bubbles Prediction"*
Candidate	Rushil R. Malode
Supervisors	K. R. Neralwar, D. Colombo
Period	June 2024 – July 2024
Internship	*"Identification of Stellar Feedback features in STARFORGE Simulation"*
Candidate	Suryansh Patidar
Supervisors	D. Colombo, K. R. Neralwar
Period	May 2024 – June 2024

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Experience

Internship

10/2019–11/2019 **Intern**, *Dr. Dario Colombo*, MPIfR, Bonn, Germany
Studying the relationship between the molecular gas distribution of the SEDIGISM survey and the stellar models of GAIA to investigate the Galactic warp.

05/2019–07/2019 **Intern**, *Dr. Benjamin Magnelli*, AIfA, Bonn, Germany
Data reduction and analysis of the ALMA Band 3 data for AzTEC galaxies using the Common Astronomy Software Application (CASA).

05/2016–08/2016 **Summer Intern**, *Dr. K. Sasikumar Raja*, IISER, Pune, India
Design and characterisation of a low-frequency log-periodic dipole antenna (LPDA) to monitor radio transient emissions from the solar corona using high-frequency structure simulator (HFSS).

Scientific organising committee

09/2024 Workshop on ML II: Exploring Machine Learning in Astrophysics, Cologne, Germany

06/2024 Workshop on ML I: Beginner's Guide to Deep Learning, Cologne, Germany

05/2023 International max planck research school (IMPRS) conference, Bonn, Germany

Telescope observations

2021-2022 **Atacama Pathfinder EXperiment (APEX)**, Observer, Atacama, Chile
Assisted the operators at the single-dish (sub-)millimeter telescope in collecting data for multiple projects.

Outreach

2016–2018 **Member**, *Astro club*, Fergusson College, Pune, India
Organised students' seminars, exibitions and poster presentations related to astronomy as a member of Astro Club.

02/2017 & 02/2018 **Volunteer, National Science Day**, *IUCAA*, Pune, India
Presented a poster on "The Multiwavelength Observations of the Sun" (Year 1) and gave a talk on Galileo Galilei's life and work (Year 2) to the general public.

Computer skills

Programming	Python 2 & 3, BASH
Packages	tensorflow, pytorch, yt, scikit-learn, astrodendro, pandas, matplotlib, seaborn
Astrophysical Software	RADMC-3D, Common Astronomy Software Applications (CASA), Modules for Experiments in Stellar Astrophysics (MESA), Continuum and Line Analysis Single-dish Software (GILDAS-CLASS), High-frequency structure simulator (HFSS)
Others	Slurm data scheduler, wandb.ai

Collaborations

STARFORGE The "**Star Formation in Gaseous Environments**" Project is a multi-institution initiative to develop cutting-edge computer simulations of star formation, and to use them to tackle some of the biggest questions in star formation: "`users.flatironinstitute.org/~mgrudic/starforge`"

OGHReS "**Outer Galaxy High Resoulution Survey**" is a 100 deg^2 APEX legacy survey of the outer Galactic disc in the 2-1 transitions of CO, ^{13}CO(2-1) and C^{18}O(2-1) in the 1 mm window. "`sedigism.mpifr-bonn.mpg.de/oghres/`"

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SEDIGISM "**Structure, Excitation and Dynamics of the Inner Galactic Interstellar Medium**" is a 84 deg^2 spectroscopic survey of the inner Galactic disc in the 2-1 transitions of ^{13}CO(2-1) and C^{18}O(2-1) observed using the APEX telescope. "`sedigism.mpifr-bonn.mpg.de`"

Current memberships

SFB 1601 "**Habitats of Massive Stars across Cosmic Time**" is a collaborative research center based at the University of Cologne, Germany. It focuses on studying massive stars adn their environments across cosmic time. I am part of the Project B2 (since February 2025): Studying stellar feedback in the Milky Way and Magellanic Clouds: "`sfb1601.astro.uni-koeln.de/projects/projectb/b2/`"

ASI "**Astronomical Society of India**" promotes Astronomy and related branches of science in India. It organises scientific meetings and supports the popularisation of Astronomy and other similar activities. I am a life member of ASI since January 2025: "`astron-soc.in`"

COST "**European Cooperation in Science and Technology**" is a funding organisation for research and innovation networks. I am a working group member since January 2024 : "`www.cost.eu/`"

Presentations and colloquium

02/2026 Colloquium talks on **Tracing the effects of stellar feedback on molecular gas from simulations to observations** at National Centre For Nuclear Research, Warsaw, Poland and Nicolaus Copernicus University, Torun, Poland.

06/2025 Colloquium talk on **A multiscale study of molecular gas in STARFORGE** at University of Cologne, Germany.

08/2024 Colloquia talks on **Molecular cloud morphologies and stellar feedback** at Indian Institute of Space Science and Technology, Thiruvananthapuram, India and Indian Institutes of Science Education and Research, Tirupati, India.

07/2023 Oral presentation on **Identification of stellar wind signatures in the Milky Way using the CASI-3D CNN algorithm** and poster on **Cloud morphologies and feedback in the inner Galaxy** at National Astronomical Meetings, Cardiff, UK.

07/2023 Posters on **A machine oriented hunt for feedback in Milky Way** and **Identification of stellar wind signatures in the Milky Way using the CASI-3D CNN algorithm** at the European Astronomical Society meeting, Krakow, Poland.

06/2022 Poster on **A machine oriented hunt for feedback in Milky Way** at From Stars to Galaxies II, Sweden.

02/2020 Poster on **Morphological classification of molecular clouds** at APEX 2020 meeting, Schloss Ringberg, Tegernsee, Kreuth.

06/2019 Paper presentation on **An intuitive 3D map of the Galactic warp's precession traced by classical Cepheids** at AIfA, Bonn, Germany.

02/2017 Poster on **The multiwavelength observations of Sun** at Inter-University Centre for Astronomy and Astrophysics (IUCAA) on National Science Day.

02/2016 Poster on **Maunder Minimum** at the National Conference 'Frontiers in Physics' at Pune.

Professional Development and Training

10/2025 **Basic AI Training: Secure Use of AI**, *Max-Planck-Gesellschaft*, Germany, (attended online)

08/2023 **Carl-Zeiss-Stiftung-Summer-School 2023** , *Heidelberg*, Germany

05/2023 **AI for Science - Bootcamp with NVIDIA**, *Max Planck Computation and Data Facilities*, Germany, (attended online)

01/2023 **Machine learning in Python with scikit-learn**, *FUN-MOOC*, France Université Numérique, Online course

11/2022 **Python for HPC workshop**, *Max Planck Computation and Data Facilities*, Germany, (attended online)

10/2022 **ASTRO HACK WEEK 2022 workshop**, *Heidelberg*, Germany, (attended online)

11/2021 **IAA Severo Ochoa Advanced School on Star Formation**, Granada, Spain

08/2017 **Python Programming: A Concise Introduction**, *Coursera*, Wesleyan University, Online course

05/2017 **Programming for Everybody (Getting Started with Python)**, *Coursera*, University of Michigan, Online course

04/2017 **Confronting The Big Questions: Highlights of Modern Astronomy**, *Coursera*, University of Rochester, Online course

Scientific visit

02/2026 **National Center for Nuclear Research, Warsaw and Nicolaus Copernicus University, Torun, Poland**, *1 week*

Observing programs

- Co-I of the APEX observing program M9550A_117, "Overlooked and understudied GMC G345: opportunity for a complete census of star formation with AMKIDs" (PI: S. Neupane)
- Co-I of the APEX observing program M9513C_113, "Shaping the ISM: filaments and bubbles in the outer Galaxy" (PI: D. Colombo)

Languages

- Marathi (mother-tongue)
- English
- Hindi
- German (B1 telc)

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